SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	**Shaik Fazul**	**Reg. No.:**	**192411063**
Section / Batch:	**BE.CSE**	**Date:**	**13/10/2026**

Code of Conduct

I _____Shaik Fazul (192411063)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO1-AT2

Date : 13/10/2026 **Reg No: 192411063**

ITA0329 **Name: Shaik Fazul**

Q2. Channels allocated to one cell

Channels per cell = Total channels ÷ Cluster size

560 ÷ 7 = 80

Answer: 80 channels per cell

Q3. Total simultaneous GSM voice channels

Total voice channels = Number of carriers × Time slots per carrier

100 × 8 = 800

Answer: 800 voice channels

Q4. Power ratio in decibels

Ratio (dB) = $10 \log_{10}$(Base station power ÷ Handset power)

$10 \log_{10}(80 \div 4) = 10 \log_{10}(20)$

10 × 1.3010 = 13.01 dB

Answer: 13.01 dB

Q5. Carrier-to-interference

C/I (dB) = Carrier power (dBm) − Interference power (dBm)

−60 + 92 = 32 dB

Answer: 32 dB

Q8. Power needed to double cell radius

Power factor = (New radius ÷ Original radius)²

$(2R \div R)^2 = 2^2 = 4$

Increase transmit power by a factor of 4

Q9. Total channels in a cluster

Cluster channels = Number of cells × Channels per cell

4 × 15 = 60

Answer: 60 channels

Q11. Duration of one TDMA time slot

Slot duration = Frame duration ÷ Number of slots

5 ÷ 8 = 0.625 ms

Answer: 0.625 ms

Q12. Number of GSM carrier frequencies

Convert bandwidth: 24 MHz = 24,000 kHz
Number of carriers = Total bandwidth ÷ Carrier spacing
24,000 ÷ 200 = 120
Answer: 120 carrier frequencies

Q15. Total call-handling capacity

Total capacity = Number of cells × Calls per cell
6 × 250 = 1,500
Answer: 1,500 simultaneous calls

Q17. Total bandwidth consumed

Total bandwidth = Number of channels × Bandwidth per channel
3,750 kHz ÷ 1,000 = 3.75 MHz
Answer: 3.75 MHz

Q18. Total guard-band bandwidth

Number of gaps between 30 adjacent bands = 30 − 1 = 29
Total guard bandwidth = Number of gaps × Guard band per gap
29 × 40 kHz = 1,160 kHz
1,160 kHz = 1.16 MHz
Answer: 1.16 MHz

Q20. Total network user capacity

Users = Regional centers × Base stations per center × Users per station
8 × 40 = 320; 320 × 250 = 80,000
Answer: 80,000 users

Q22. Number of FDMA channels

Convert bandwidth: 6 MHz = 6,000 kHz
Number of channels = Total spectrum ÷ Bandwidth per channel
6,000 ÷ 30 = 200
Answer: 200 channels

Q23. Offered traffic in Erlangs

Formula: Traffic = Subscribers × Calls per hour × Call duration (in hours)
Convert call duration: 2.5 minutes ÷ 60 = 2.5/60 hours
1,200 × 4 × 2.5 ÷ 60 = 200
Answer: 200 Erlangs

Q24. Maximum simultaneous users

Convert bandwidth: 40 MHz = 40,000 kHz
Users = Total bandwidth ÷ Bandwidth per user
40,000 ÷ 400 = 100
Answer: 100 simultaneous users

Q25. Total base-station user capacity

Formula: Total capacity = Number of base stations × Users per station
1,200 × 600 = 720,000
Answer: 720,000 simultaneous users

Q28. Number of roaming subscribers

Roaming subscribers = Total subscribers × Roaming percentage ÷ 100
600,000 × 0.15 = 90,000
Answer: 90,000 subscribers

Q29. Switching center utilization

Utilization (%) = Actual load ÷ Maximum capacity × 100
0.75 × 100 = 75%
Answer: 75% utilization

Q31. Number of GSM channels

Convert bandwidth: 30 MHz = 30,000 kHz
Number of channels = Total bandwidth ÷ Channel spacing
30,000 ÷ 200 = 150
Answer: 150 channels

Q32. Successful handoffs

Successful handoffs = Total attempts × Success rate ÷ 100
2,500 × 0.99 = 2,475
Answer: 2,475 successful handoffs

Q34. Total cluster user capacity

Cluster capacity = Number of cells × Users per cell
7 × 900 = 6,300
Answer: 6,300 simultaneous users

Q35. Duration of one TDMA slot

Slot duration = Frame duration ÷ Number of slots
4.8 ÷ 8 = 0.6 ms
Answer: 0.6 ms

Q38. Total cluster user capacity

Cluster capacity = Number of cells × Users per cell
9 × 600 = 5,400
Answer: 5,400 users

Q41. Total geographic coverage

Total area = Area per base station × Number of stations
2.5 × 150 = 375
Answer: 375 km²

Q45. Signal-to-noise ratio (SNR)

SNR (dB) = Signal power (dBm) − Noise power (dBm)

−85 + 115 = 30 dB

Answer: 30 dB